

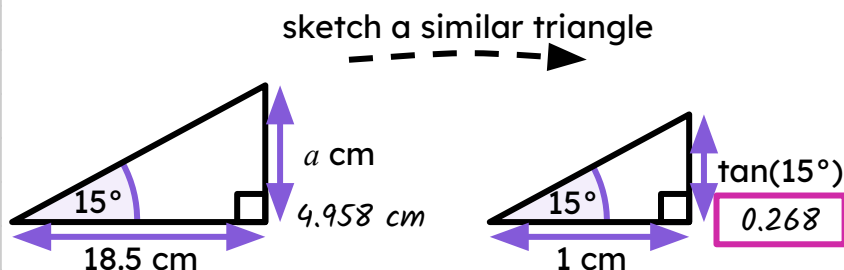


Tangent ratio

Task A: Ratio tables, the unit circle, and tangents

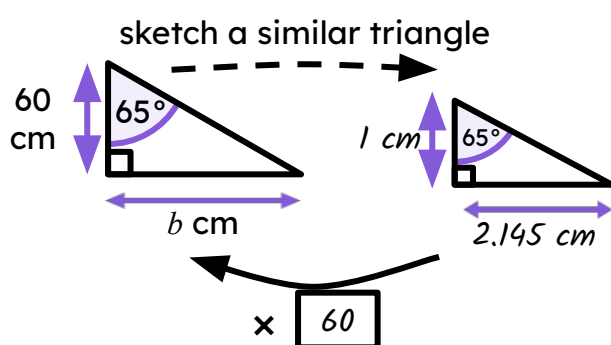
angle	sine	cosine	tangent
0	0	1	0
5	0.087	0.996	0.087
10	0.174	0.985	0.176
15	0.259	0.966	0.268
20	0.342	0.94	0.364
25	0.423	0.906	0.466
30	0.5	0.866	0.577
35	0.574	0.819	0.7
40	0.643	0.766	0.839
45	0.707	0.707	1
50	0.766	0.643	1.192
55	0.819	0.574	1.428
60	0.866	0.5	1.732
65	0.906	0.423	2.145
70	0.94	0.342	2.747
75	0.966	0.259	3.732
80	0.985	0.174	5.671
85	0.996	0.087	11.43
90	1	0	∞

1) Fill in the blanks and ratio table. By considering corresponding sides, find the length of side labelled a cm.



	adjacent	opposite
	1	0.268
$\times 18.5$	18.5	4.958

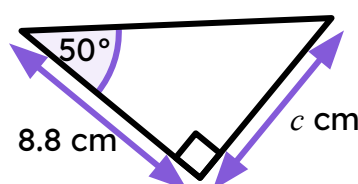
2) By sketching similar triangles with a adjacent of 1 unit, find the length of the side indicated on each triangle.



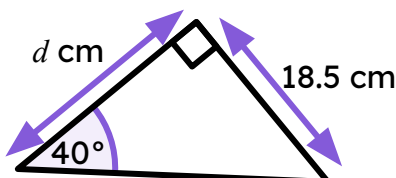
$$b = 60 \times 2.145$$

$$= 128.7$$

The length of the side marked b is 128.7 cm



$$c = 1.04896$$



$$d = 22.05$$

Task B: The opposite and the adjacent

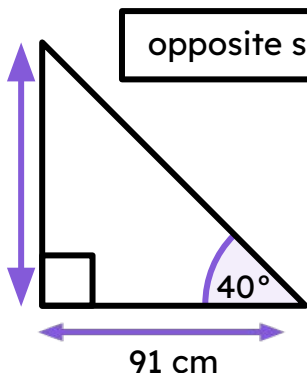
1) Find the length of the side opposite the labelled angle using this relationship.

a)

$$\text{adj} \times \tan(\theta^\circ) \text{ cm}$$

$$91 \times \tan(40^\circ) \text{ cm}$$

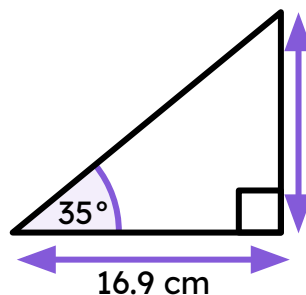
$$91 \times 0.839 \text{ cm} \\ = 76.349 \text{ cm}$$



$$\text{opposite side} = \text{adj} \times \tan(\theta^\circ)$$

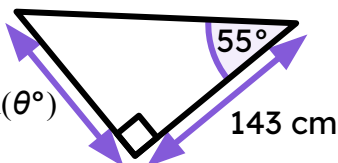
b)

$$16.9 \times \tan(35^\circ) \text{ cm} \\ = 11.83 \text{ cm}$$



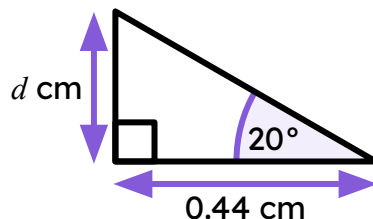
c)

$$\text{adj} \times \tan(\theta^\circ) \\ = 204.204 \text{ cm}$$



d)

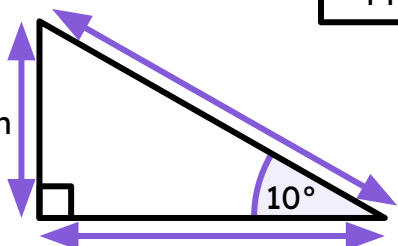
$$d \text{ cm} \\ d = 0.16016 \text{ cm}$$



2) Find the length of the adjacent when given the side opposite the labelled angle.

a)

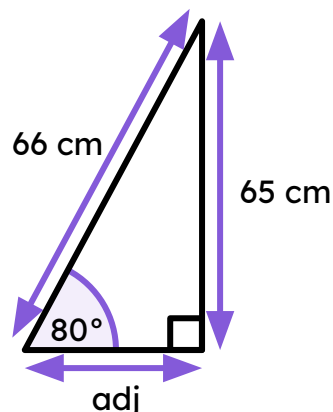
$$\text{adj} = \frac{\text{opp}}{0.176} = 500 \text{ cm}$$



$$\text{opposite side} = \text{adj} \times \tan(\theta^\circ)$$

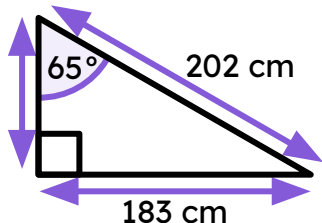
b)

$$\text{adj} = \frac{65}{5.671} \\ = 11.46 \text{ cm}$$



c)

$$\text{adj} = \frac{202}{2.37} = 85.31 \text{ cm}$$



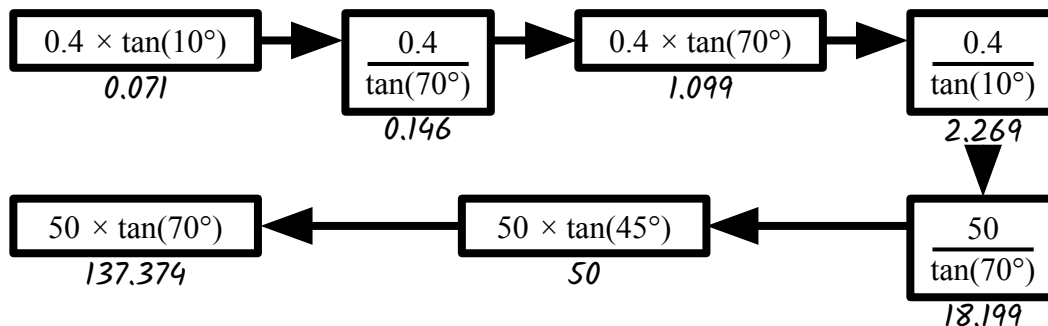
d) A right-angled triangle has an angle of 85°. The length of the side opposite this angle is 142 cm.

$$= 12.42 \text{ cm}$$



Task C: Effectively using a calculator for the tan ratio

1) Starting with the smallest, put these side lengths of right-angled triangles in order of size.



2) Find the lengths of all sides in each of these triangles (3 d.p.)

